1. Looking for the hidden treasure …

In a room of *M x N* cells there exists a particular position where the most delicious food is present, and we are going to find out using PSO.
You are a particle of the swarm, and you work together with your classmates looking for the food.
You are given the initial positions and velocities, as well as, the three parameters:

w=0.8: inertia
cp=0.7: influence of the particular best position
cg=0.8: influence of the global best position

You must calculate your new position and velocity in each iteration, according to the following equations and keep the track of your particular best evaluation as well as the global best evaluation.
The modification of your position can be mathematically modelled according to the following equation:

$$V_i^{k+1} = wV_i^k + c_p\, r_p(S_{pbest_i} - S_i^k) + c_g\, r_g(S_{gbest} - S_i^k)$$
$$S_i^{k+1} = S_i^k + V_i^{k+1}$$

where,

V_i^k : velocity of agent i at iteration k,
w: influence of the previous velocity,
c_j : influence of the global or local previous solution,
r_j : uniformly distributed random number between 0 and 1,
S_i^k : current position of agent i at iteration k,
$pbest_i$: pbest of agent i,
gbest: gbest of the group.

w=0.8: inertia
cp=0.7: influence of the particular best position
cg=0.8: influence of the global best position

k=0 gbest=				
S^0		feval	V^0	Spbest
22.2135	12.0427	1.0643	0 0	
26.1881	19.7741	9.7224	0 0	
20.9158	10.7442	1.1140	0 0	
22.7604	12.3982	1.5916	0 0	
21.8738	10.3611	0.6512	0 0	
23.7841	12.8367	2.5606	0 0	
17.6520	9.2159	4.6998	0 0	
25.5818	8.4024	4.4246	0 0	
34.4546	9.6090	12.5320	0 0	
24.6263	10.4131	2.6911	0 0	

	k=1 gbest =				
r_p	r_g	V^1	S^1	feval	pbest
0.9353	0.3402				
0.5746	0.1398				
0.3187	0.8329				
0.5457	0.9399				
0.8452	0.5252				
0.3789	0.9216				
0.0152	0.0623				
0.7711	0.3040				
0.0199	0.5987				
0.5190	0.1252				
0.9353	0.3402				